SdS

Day-3

→ Types of Neural Networks

1) Multi-Layer Perceptrons (MLP) (ANN)
- input layer → h1 → h2 → output layer
- Supervised ML problem.

2) Convolutional Neural Network (CNN)
- it is a special NN where at least one Convolutional layer used.
- used in image & video processing.

3) Recurrent Neural Network (RNN)

4) Auto Encoders
- used for compressed i/p data without loosing information

5) Generative Adversarial Networks (GAN)
used to generate (images photo, video & Music

→ History of Deep learning

Ch-1) Start in 1960's

Ch-2) 1969. The first AI winter
perceptron can't learn XOR fn.

Ch-3) 1980's - The rise
learning representations using Back-propogating errors perspective.

SDS

Ch-4) The 90's. The second AI winter
- Label data
- Communication power GW
- other algorithm came SVM, Random forest

Ch-5) 2006
- unsupervised pre-training papers published
- Hence the term Deep Learning

Ch-6) 2012. The final Rise
- Imagenet Challenge
- Deep Learning on GPU

→ Application of Deep Learning

1) Self-Driving Cars
2) Game playing Agent
3) Virtual Assistances
4) Image Colorization
5) Adding audio to Mute Videos
6) Image Caption Generation
7) Text translation
8) Pixel Restoration
9) Object Detection
10) GAN
11) Deep Dreaming